

Full Length Article

Heterogeneities dominate mechanical performance of additively manufactured metal lattice struts

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ABSTRACT

Architected structural metamaterials, also known as lattice, truss, or acoustic materials, provide opportunities to produce tailored effective properties that are not achievable in bulk monolithic materials. These topologies are typically designed under the assumption of uniform, isotropic base material properties taken from reference databases and without consideration for sub-optimal as-printed properties or off-nominal dimensional heterogeneities. However, manufacturing imperfections such as surface roughness are present throughout the lattices and their constituent struts create significant variability in mechanical properties and part performance. This study utilized a customized tensile bar with a gauge section consisting of five parallel struts loaded in a stretch (tensile) orientation to examine the impact of manufacturing heterogeneities on quasi-static deformation of the struts, with a focus on ultimate tensile strength and ductility. The customized tensile specimen was designed to prevent damage during handling, despite the sub-millimeter thickness of each strut, and to enable efficient, high-throughput mechanical testing. The strut tensile specimens and reference monolithic tensile bars were manufactured using a direct metal laser sintering (also known as laser powder bed fusion or selective laser melting) process in a precipitation hardened stainless steel alloy, 17-4PH, with minimum feature sizes ranging from 0.5–0.82 mm, comparable to minimum allowable dimensions for the process. Over 70 tensile stress-strain tests were performed revealing that the effective mechanical properties of the struts were highly stochastic, considerably inferior to the properties of larger as-printed reference tensile bars, and well below the minimum allowable values for the alloy. Pre- and post-test non-destructive analyses revealed that the primary source of the reduced properties and increased variability was attributable to heterogeneous surface topography with stress-concentrating contours and commensurate reduction in effective load-bearing area.

1. Introduction

Additive manufacturing, AM, provides unique manufacturing capabilities over traditional fabrication methods. Traditional, or subtractive, methods require tool access paths to create the part while AM does not require tool access. This allows designers to create complex geometries with internal features by using additive manufacturing. One important application for this geometric flexibility is the production of lattice materials comprised of repeating unit cells throughout a volume. Lattice materials, also known as structural metamaterials [1], architected materials [2], or truss materials [3], can be more structurally efficient than foams with similar design and material [4]. They also can enable effective mechanical properties, such as negative Poisson's ratio [5,6], that are not achievable with monolithic materials. An example

lattice structure is the octet truss unit cell design, shown in Fig. 1, that provides a favorable strength-to-weight ratio [7,8]. However, the current paradigm for lattice design typically assumes idealized deterministic material properties for the base material and uniform nominal feature dimensions. The present study explores the true heterogeneities of struts, the basic building blocks for most lattice topologies, in a metal 3D printing technology using a laser powder bed process known as direct metal laser sintering (DMLS).

There are several reasons to expect that as-built lattice struts will deviate substantially from their idealized nominal state that is used in design. Most importantly, all additive manufacturing processes produce a native surface topography associated with the solidification dynamics [9]. In addition, surface roughness can depend on orientation with respect to the build direction. For example, in the powder bed process

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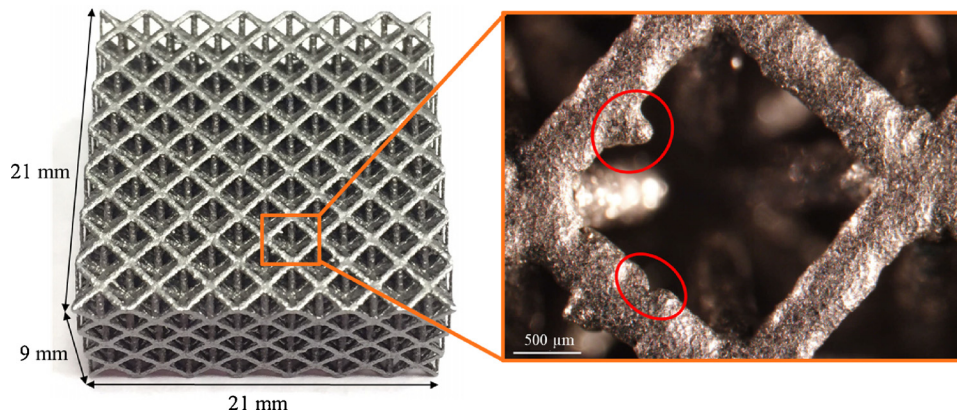


Fig. 1. Non-cylindrical struts with extra material (top-left) and missing material (bottom-left).

used in the current study, horizontal features above the build plate are supported only by loose powder, resulting in particularly poor surface finish [10]. In the example shown in Fig. 1, the lattice was designed to have cylindrical struts with a diameter of 0.5 mm, but the manufactured struts have a varying cross-sectional area and stress-concentrating features. While the present study focuses on this DMLS example, the overarching conclusion that unintended heterogeneities play a substantial role in effective lattice behavior can extend qualitatively to any AM process that is utilized to produce fine scale features.

The geometric heterogeneities not only affect the nominal performance of the lattice, such as its effective elastic response and yield behavior, but can also trigger premature structural failure. The octet truss architecture, like many lattice designs, is susceptible to localized failure cascades: when one strut fails, the neighboring struts are more highly stressed due to load redistribution and failure can easily percolate [11]. This unstable cascading failure can be difficult to observe due to the lack of visual access, requiring computed tomography [12,13] albeit with limited time resolution. While there is value in evaluating the performance of individual struts as in prior studies [14], such a single isolated strut with dimensions in the 0.5-mm regime can be susceptible to handling damage during printing and extraction from the build plate. Instead, in this study the focus was on the combined behavior of a more robust arrangement of five struts in parallel-strut tensile specimen and comparing the observed response to that expected from monolithic macroscale tensile bars. This approach allowed the use of digital image correlation to observe individual strut strains as well as visual access of the heterogeneities and their effect on the failure process. The tensile behavior of 225 struts were characterized over a range of strut sizes and build orientations, utilizing efficient testing methods to compare to statistical distributions of reference material. Prior to destructive testing, the samples were inspected using x-ray computed tomography (CT). Following tensile testing, the fracture surfaces and testing data were analyzed to increase understanding of defect impacts. All samples were manufactured through DMLS with 17-4PH stainless steel powder.

This study complements and expands the knowledge base from previous research. Gümruk and Mines [14], for example, examined the tensile behavior of individual struts and found them to exhibit reduced strength and Young's modulus relative to bulk samples, but their study was limited to a single strut size and orientation. In contrast, the present study examined the effects of strut size and build orientation on their heterogeneous effective properties and compared them to the nominal as-printed material properties exhibited by larger scale monolithic tensile bars. Weißmann et al. [15] compared the surface roughness, hardness, and compressive strength of struts of various diameters and lengths fabricated with selective laser melting (SLM) versus electron beam melting (EBM), but they confined their study to the influence of surface properties (roughness, as-built diameter) and post-processing

on compressive properties. In contrast, the present study utilized CT imaging and fractography to build a more comprehensive understanding of the impacts of defects on the mechanical properties of additively manufactured struts and lattice structures. A similar inspection procedure was applied to cast lattice struts by Zhou et al. [16] but unlike the present study, they did not investigate the unique heterogeneities associated with additively manufactured lattices.

2. Method

The 5-strut tensile design employed in the present study consisted of 5 parallel tensile bars, four at each corner and one in the center akin to the five dots on a gaming die, as shown in Fig. 2. During the macroscopic compressive deformation of conventional lattice unit cells, different struts are loaded in tension, bending, and compression, depending on their location, orientation, and local constraint. In many lattice designs, the tensile-loaded struts typically limit the catastrophic failure of the structure. This is particularly relevant in stretch-dominated lattice designs such as the octet truss [4]. By adopting the unique 5-strut tensile design, this study isolates only the tensile mode without the confounding effects of other loading modes. It also avoids the loading distortions associated with testing a specific unit cell geometry in isolation from a periodically repeating lattice structure [17]. There

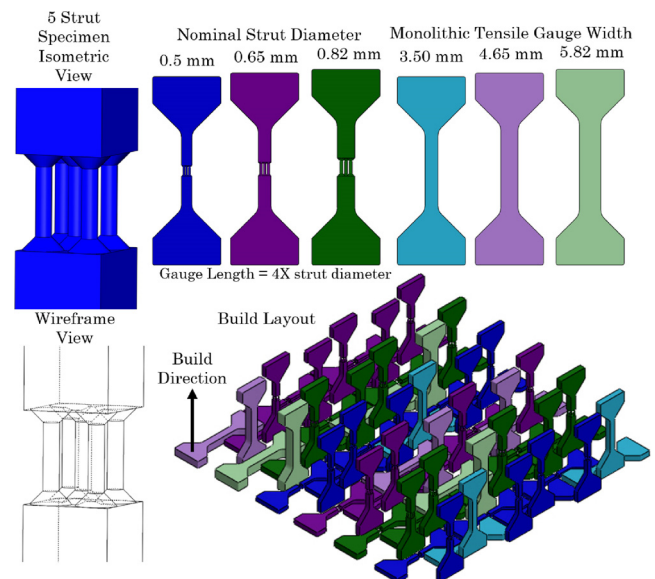


Fig. 2. Models of strut (top-left) and monolithic (top-right) samples showing the different sizes in different colours that correspond to the sample distribution in the build layout (bottom).

are two advantages to testing five struts in parallel versus testing a single strut as in [14]: (1) the 5-strut configuration is likely to be less susceptible to unintentional damage during handling, including removal from the build plate, although great care is still required to remove the 5-strut specimens from the build plate, and (2) the method provides high-throughput observations on stochastic response of struts – a total of 225 struts were tested in this study in a matter of hours—which would be burdensome if each single strut were carefully fixtured, for example, by aligning and adhering it to tabs for gripping and gages for measuring displacement. The strut tensile design was limited to 5 struts to facilitate virtual extensometer strain measurement on each strut, which requires visual line-of-sight from two DIC cameras, as described subsequently. A detraction of the current approach is that the true stress in each isolated strut is not estimated; instead only the stress of the summed five struts is analyzed as a composite response.

Both the strut samples and the monolithic samples were designed in three different sizes, as shown in Fig. 2. The strut samples were designed with nominal diameters of 0.50 mm, 0.65 mm, and 0.82 mm, shown left to right in Fig. 2. These sizes were chosen as they represented the lower end of resolvable feature sizes for the particular printing process employed in the present study. The monolithic samples were designed to have square gauge widths of 3.50 mm, 4.65 mm, and 5.82 mm. The naming convention used in this study is based on the nominal strut diameter or the width of the sample for the strut or monolithic samples, respectively.

A combination of horizontal and vertical test samples were arranged in a build layout shown in Fig. 2. The build layout was designed to effectively use the available build space and to locate each sample type in different regions of the build. Each size and orientation combination of the strut samples were replicated nine times while the monolithic samples were replicated three times.

The samples were manufactured using DMLS out of a precipitation hardenable stainless steel alloy, commonly known as 17-4PH stainless steel powder. The powder, supplied by Praxair Surface Technologies under product name TruForm 174-2 (ZFE276-02BK Lot 5), was sieved to filter all powder above 44 μm . The reported powder composition was 16.47% chromium, 4.42% nickel, 3.25% copper, 0.75% silicon, 0.18% columbium, 0.05% manganese, 0.02% carbon, 0.02% oxygen, 0.01% nitrogen, and other elements less than 0.01% balanced with iron (74.8%). The additive manufacturing of the build was performed by a commercial vendor, Stratasys Direct Manufacturing, using their EOS M270 with their standard proprietary process settings intended to produce optimal 17-4PH quality. The only deviation from standard practice was to reduce the layer thickness from 40 to 20 μm to improve feature quality. No attempt was made in this study to explore process setting variations beyond the standard EOS settings for 17-4PH powder. After manufacturing, the parts were austenitized at 1900 °F (1038 °C) for one hour while on the build plate to relieve residual stresses. After removal from the build plate, the parts underwent hot isostatic pressing (HIP) at 2125 $\pm$ 25 °F (1163 $\pm$ 14 °C) at a pressure of 14.75 $\pm$ 0.25 ksi (102 $\pm$ 1.7 MPa) for 240 $\pm$ 60 min. After the HIP process, the specimens were subsequently re-austenitized at 1900 °F (1038 °C) for an hour and cooled followed by heat treatment at 1150 $\pm$ 25 °F (621 $\pm$ 14 °C) for 1 h. An example of the microstructure after heat treatment is shown in Fig. 3. Subsequent Archimedes density tests indicated that the material was 98.4% dense. Due to manufacturability issues, two separate builds were used to harvest successfully printed test structures. Horizontal strut samples were harvested from the first build whereas vertical strut samples were harvested from the second build which utilized a recoater brush as opposed to the more conventional recoater blade. While this may add a confounding effect of build-to-build variation, monolithic tensile bars from both builds exhibited consistent mechanical properties.

After visual inspection, all samples were non-destructively analyzed using CT to observe porosity, surface topography, and as-built dimensions. The high density of additively manufactured 17-4PH steel parts

can cause x-ray scattering that impacts the radiological examination. To increase the signal-to-noise ratio and narrow the x-ray spectrum, a copper plate was placed in front of the x-ray tube. The x-ray tube was operated at 220 kV to collect more than 1000 projections over 1–2 hours. Volume Graphics was used to reconstruct the sample surface and internal geometry with a spatial resolution of 11 μm . To quantify the strut's surface roughness, slices of CT data were analyzed using the image processing toolbox in MATLAB. Two longitudinal slices of each strut were analyzed for a total of four roughness measurements per strut (twenty measurements per sample). MATLAB identified the distance between the edge of the strut and the midline to calculate the roughness through the root-mean-square method. Similarly, cross-sectional slices were analyzed to determine the cross-sectional area throughout the gauge section of all samples. Cross-sectional slices were analyzed at 11 μm intervals throughout the entire gauge section. To understand the role of surface topography on tensile performance, the scanned geometries were segmented and structural analysis was performed using VGSTUDIO Max 3.1 made by Volume Graphics. Two steps were used to segment each strut sample from its scan volume: an initial ISO-thresholding was used to find the initial contour and then an active-contour algorithm [18] was used to improve the accuracy of the segmentation. After segmentation was complete, two regions of interest (ROI) were identified such that each encompassed approximately half of the monolithic gauge region of the strut samples at opposing ends. As such, the regions of the fixed and loaded boundary conditions of the structural mechanics simulation were defined. Linear elastic finite element simulations were performed using a meshless discretization of the segmented geometry based on the immersed boundary method [19] with linear basis functions and a conjugate gradient solver. The method was validated in separate studies [20] and compared against a conventional commercially available finite element package. Using the voxels inherent to the data set, a simulation cell size of 4 voxels (44 μm) was found to provide a suitable balance between computational efficiency, convergence and spatial resolution. A detailed convergence study was not performed as the purpose of the finite element simulation was to provide only approximate first-order semi-quantitative insight into the role of surface topography on stress redistribution. Accurate determination of local stresses would require a more robust analysis.

Following non-destructive analysis, each sample was loaded on a MTS servo-hydraulic load frame with a 22 kN load cell and/or a 156 kN load cell depending on sample size. Actuator rates controlled each test at a strain rate of approximately 0.0016 s^{-1} corresponding to test durations of ~1–2 minutes. Throughout the tensile tests, data was acquired by two parallel and independent acquisition systems. The MTS control software acquired load and displacement data at 15 Hz. In parallel, Correlated Solutions Vic-Snap software stored camera frame captures at a rate of 1 Hz along with a log file of analog voltage values corresponding to the load and displacement values at each frame capture. The 1 Hz frame captures were obtained using two Point Grey Grasshopper cameras (model GS3-UR-41C5M-C) each with a Navitar 12x zoom lens (~2 mm field of view) positioned to view the full height of the struts. The two cameras were not used for 3D digital image correlation, but rather to view both the front and back of the specimen simultaneously, permitting 2D digital image correlation on the two front and two back struts. A virtual extensometer strain measurement was performed using digital image correlation (DIC) using commercial software from Correlated Solutions, Inc. The gauge length for the virtual extensometer was the entire length of the actual gauge section, 4X the strut diameter or gauge width. Sequential images taken before the beginning of the test were used to estimate the strain noise floor, which was estimated to be 50 $\mu\epsilon$ ($\pm$ 1 standard deviation), sufficiently low to not affect the interpretation of results in this study. In the case of the 5-strut specimens, strain was measured on only one of the five struts. Since all five struts were attached to the same much larger monolithic sections, this approximately rigid boundary condition meant that strain measurements on one strut were a reasonable proxy for strain in all five

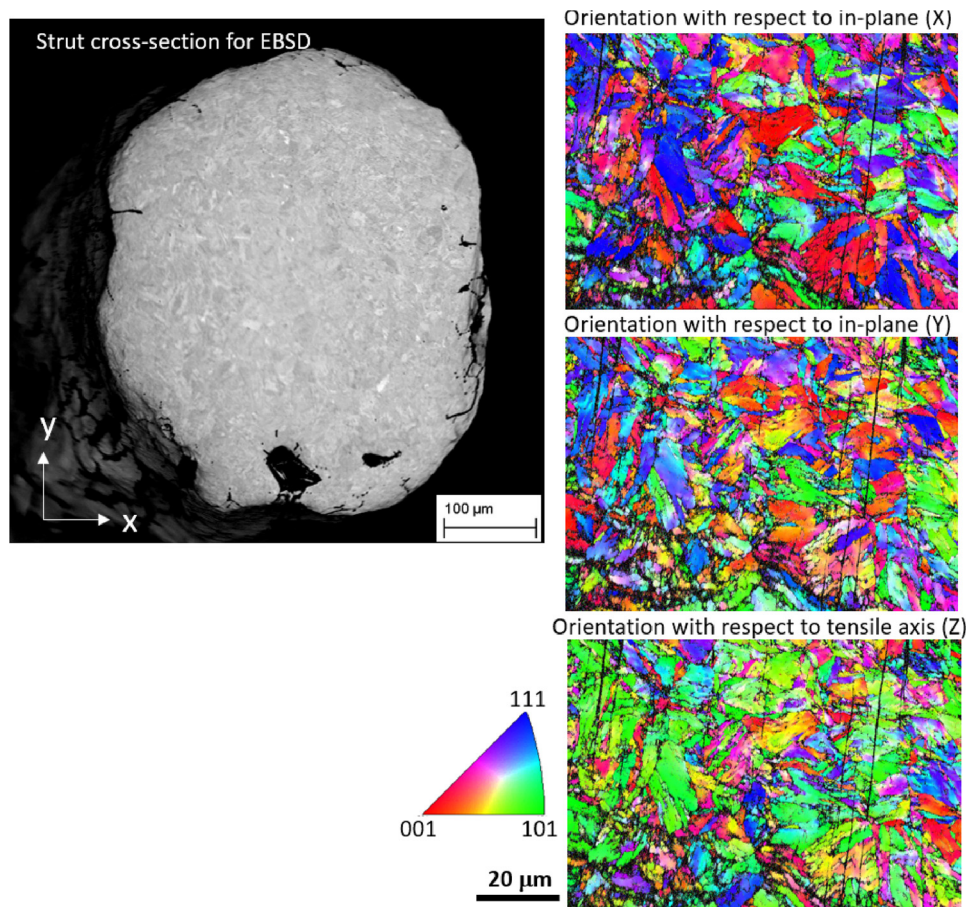


Fig. 3. The microstructure after heat-treatment in a vertical strut as characterized by electron backscatter diffraction (EBSD). Only approximately 2% of the image indexed as FCC (retained austenite), with the remainder indexing as BCC due to the low tetragonal distortion in the martensite phase. Notably, the grain size was much finer than the strut dimensions and there was little in-plane crystallographic texture although a (110) preference along the tensile axis.

struts. Post-test DIC analysis of all struts confirmed that the strain discrepancy between struts was $< 1\%$, even after the reference strut had broken. The native surface roughness of the specimens provided the needed speckle contrast for digital image correlation. For the 5-strut specimens, the cross-sectional area required for calculating engineering stress was estimated as the sum of the minimum cross-sectional area for each of the five struts. Fractography was imaged post-mortem using a JEOL JSM IT-300HR scanning electron microscope at accelerating voltages of 10 and 30 keV.

3. Results

3.1. Strut quality

After removal from the build plate and heat treatment, all test samples were CT scanned to quantify internal porosity, surface topography, and deviation from designed dimensions. For the strut samples, CT data revealed substantial deviations from the nominal shape and dimension as shown in Fig. 4. This figure illustrates that the vertical struts were generally more cylindrical with less significant topological defects compared to the horizontal struts, an effect that is quantified in the Discussion section. This finding is consistent with previous research that documents the deterioration of strut geometry with decreasing inclination angle relative to the build platform [17,21].

To approximate the actual cross-sectional area for more accurate stress determination, a MATLAB algorithm extracted the minimum cross-sectional area for each strut based on more than 200 CT slices across the gauge section. These minimum values were averaged for all struts in a given population, as is reported in Table 1.

3.2. Stress-strain response

Following non-destructive evaluation, all samples were tensile tested to failure. Fig. 5 displays the stress-strain response of the reference monolithic tensile bars delineated by size and orientation. As discussed in the previous section, the actual as-printed cross-sectional areas deviated from the designed dimensions, which impacted the apparent stress response. For this reason, two stress-strain plots compare stress calculated using the nominal as-designed dimension and using the actual as-printed dimensions. Accordingly, the terminology “nominal” and “actual” is used throughout the remainder of the paper to distinguish these two methods. Of the 27 monolithic tests, five failed at anomalously low strain (less than 0.015) during the elastic loading ramp prior to substantial yielding. These anomalous “low ductility” results are treated in more detail in the next sub-section.

Stress-strain curves for 5-strut specimens are displayed in Fig. 6, categorized by strut diameter and build orientation. In the case of the strut samples, the stress values were calculated by dividing the applied force by the sum of the minimum cross-sectional areas for all five individual strut members in the specimen. The stress calculations used for stress-strain curves and statistical analysis assume that the stress is homogenous and uniaxial throughout the gauge section. A more detailed interpretation requires the analysis of the gradient multiaxial stresses that arise due to the 3-dimensional surface topography, as is described in Section 4. The macroscopic strut strain was measured on one of the five struts using DIC, and assumed to be equal in all five struts, an assumption that was confirmed to be reasonable due to use of displacement control and the stiff (essentially rigid) attachment of the struts to the monolithic section of the tensile bar. As expected, the struts did not fail on a single plane but at heterogeneous locations across the gauge section, as shown in Fig. 7. The failure locations shown in Fig. 7

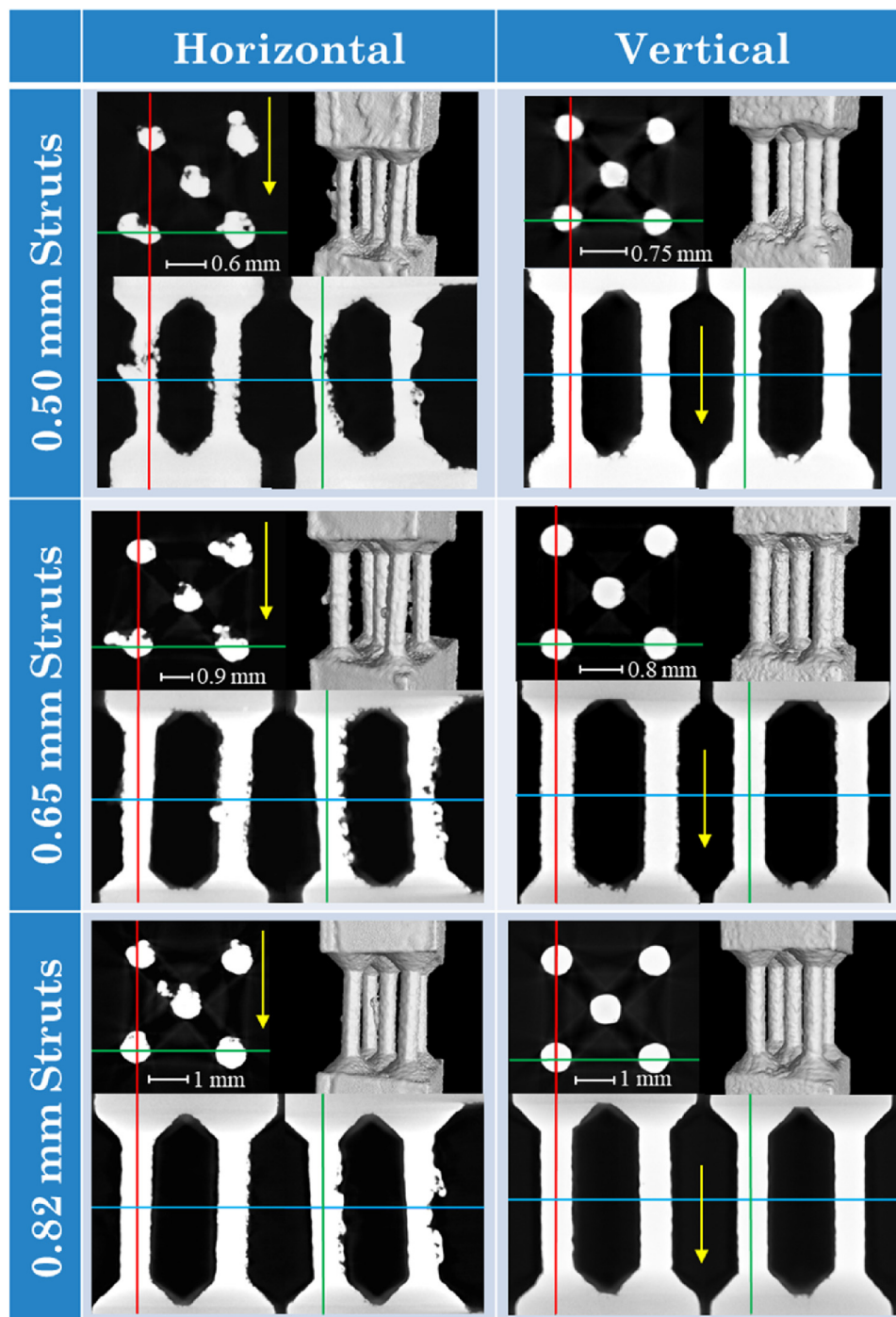


Fig. 4. Representative strut sample CT data for all sample configurations. Yellow arrows indicate the build direction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Table 1

Comparison of nominal strut areas to the actual as-printed minimum strut areas.

Nominal strut Diameter	Nominal Strut Area (mm ²)	Horizontal			Vertical		
		Number of 5-Strut Samples Tested	Average of Minimum Area (mm ²)	Standard Deviation of Minimum Area (mm ²)	Number of 5-Strut Samples Tested	Average of Minimum Area (mm ²)	Standard Deviation of Minimum Area (mm ²)
0.5 mm struts	0.196	5	0.113	0.034	6	0.115	0.009
0.65 mm struts	0.332	8	0.222	0.055	9	0.284	0.013
0.82 mm struts	0.528	8	0.372	0.071	9	0.480	0.022

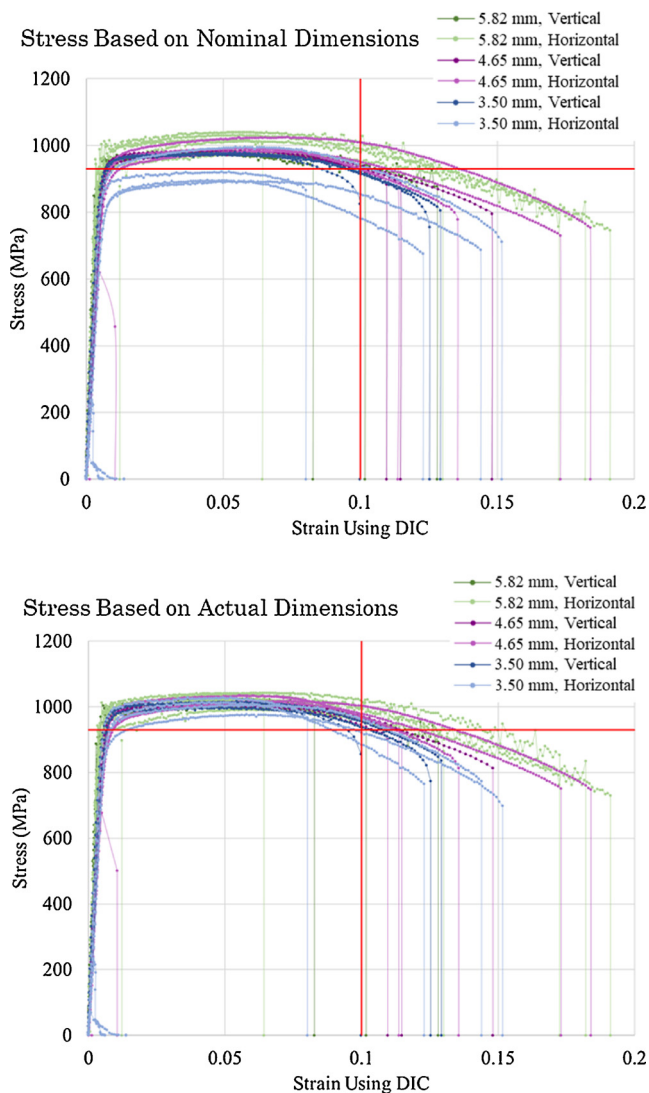


Fig. 5. Stress-strain responses for monolithic samples comparing stresses calculated from nominal as-designed areas (top) and actual as-built areas (bottom). The red lines represent the minimum allowable ultimate strength and strain for 17-4PH as per AMS 5604. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

did not always correspond to the of minimum cross-sectional area as determined by CT, likely because the local stresses driving deformation were complex due to the surface topography, as described in Section 4.2.

An alternative way of displaying the tensile data is to plot force versus cross-head displacement (normalized gauge length), as shown for representative curves in Fig. 8. These smoother response curves (due to higher data density and higher resolution force data) illustrate the instantaneous load drop associated with the fracture of each of the five struts. However, the x-axis is not a direct measure of material strain due to contributions from load frame compliance and specimen settling in the grips. In addition to these four basic examples, a fifth result in Fig. 8 shows an example of one of the five monolithic tests that exhibited anomalously low ductility, failing in the elastic region before substantial macroscopic yielding had occurred.

In the case of the reference monolithic tensile bars, specimen orientation and size did not significantly affect the mechanical properties, once corrected for actual sample dimensions and excluding low-ductility outliers. The average and standard deviation of ultimate tensile

strengths are reported in Table 2. For the monolithic samples, the actual ultimate strengths had less deviation than the nominal ultimate strengths, illustrating the important effect of dimensional variability. Setting aside the low ductility outliers, all other monolithic tensile bars had an ultimate strength above the minimum allowable value of 930 MPa from AMS 5604. However, even when censoring the clearly anomalous samples, three of the remaining 22 tests exhibited failure strains below the minimum allowable value of 10%. With the exception of the low-ductility samples, the horizontal and vertical monolithic samples exhibited similar stress-strain curves.

The strut specimens were analyzed in the context of failure strain. In this analysis, we analyzed the variability of the final (5th) strut failure, as if this structure was to be designed against final catastrophic failure. The corresponding “strain at failure” rupture distributions for the 5th strut are shown in Table 3. For some applications, the first strut failure or the total energy dissipated may be more important metrics, although that data is not presented here.

There are features of the strut stress-strain curve that warrant explanation. Unlike the monolithic samples, the stress-strain curves for strut samples have multiple stress drops past peak load as each of the five struts broke in sequence. Generally, the ultimate strength for horizontal strut samples was significantly lower than the ultimate strength for vertical samples, and both were lower than the reference monolithic behavior, even after accounting for actual sample dimensions. In the best-case scenario, the actual ultimate strength of vertical struts was within 10% of the reference monolithic samples. With regard to failure strain, the average final failure of the strut tensile specimens was also below that of the reference monolithic samples. Here again, while the horizontal struts deviated substantially from the monolithic behavior, the vertical struts were < 10% below their monolithic counterparts. However, this result only describes the final failure of the fifth strut in the strut specimen. As is evident in Fig. 8, the first strut failure occurred at a much lower strain value.

3.3. Low-ductility monolithic samples

Five horizontal monolithic samples failed anomalously before the onset of macroscopic yielding. Visual inspection of the fracture surface of the five low-ductility samples revealed obvious brown-orange tinted discolouration, as shown in Fig. 9. The amount of discolouration on the fracture surface ranged from the entire fracture surface to a small corner of the surface. There appeared to be an inverse correlation between the amount of discolouration and the part performance (strength and strain at failure). CT data from two of the five low-ductility samples showed large pockets of networked, interconnected porosity near the failure location. The other three samples fractured close to the grip, which was outside the range of the CT scans. Fig. 9 shows a typical example of fractographic and CT inspection from a sample that failed at a nominal stress of 699 MPa. The discoloured region in the upper left of the fracture surface matches the pattern of CT porosity in the sample.

A previous study by Boyce et al. revealed a similar anomalous failure mode [22]. The study attributed low-ductility behavior to interconnected pores that connect to the free surface of the part; the interconnected network of pores most likely permitted the interior of the sample to oxidize during heat treatment [22]. While the crack-like geometry of the networked pores is thought to drive the failure process, it remains unclear if the chemical oxidation also degrades the material resistance to fracture [22]. As in the previous study, the test samples were heat treated in air potentially causing oxidation throughout the pore network. The parts underwent a HIP treatment between the two heat treatments to reduce porosity. However, the parts were not canned during HIP so the treatment did not impact the pores connected to the surface. Canning the parts would have created the pressure differential necessary to collapse those pores.

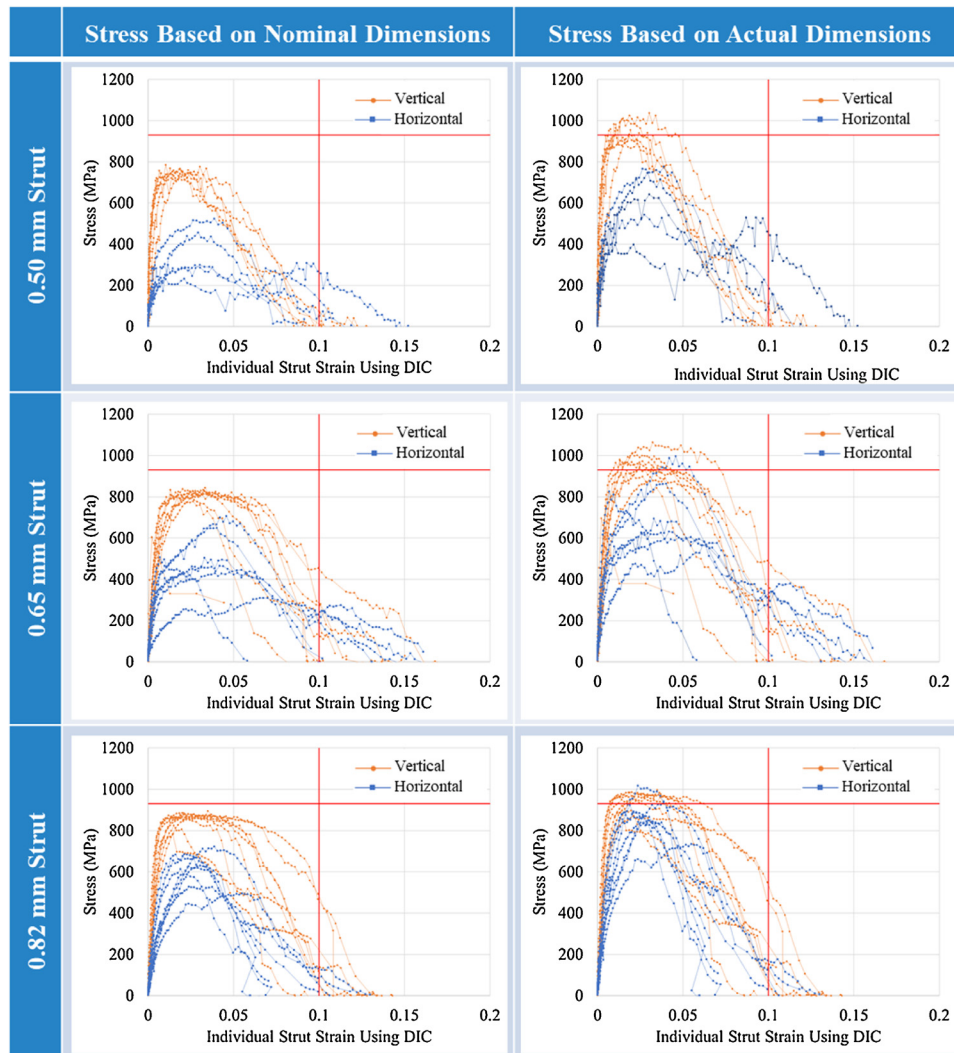


Fig. 6. Stress-strain responses for strut samples separated by strut size comparing stresses calculated from as-designed areas (left) and as-built areas (right). The red lines represent the minimum allowable ultimate strength and strain for 17-4PH as per AMS 5604. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

3.4. Strut fractography

After tensile testing, a scanning electron microscope (SEM) was used to examine the fracture surfaces. SEM inspection revealed the heterogeneities that caused deviations in strut performance from the monolithic behavior. Three significant factors were of interest in fractographic inspections: the effect of surface topography both as a source of stress concentration and as a reduction in the true load bearing area, internal defects, and interconnected porosity that tunneled to the free surface.

Fig. 10 presents representative images of the typical macroscopic morphology of fractured struts, showing clear deviation from nominal circular cross-section due to the native surface roughness of the printing process. The visually significant contrast in build quality between horizontal and vertical build orientations is in agreement with CT imaging results presented in Fig. 4, and shows the reduction in effective

cross-sectional area that contributed to the wide variation of performance results for horizontal samples. Additionally, surface roughness had a greater effect as the strut diameter decreased, as shown by SEM images in Fig. 10. Higher magnification images of typical fracture surface defects are shown in Fig. 11. The dripping surface roughness defect was only present in the horizontal struts and helps to explain the reduced mechanical properties associated with these struts. The example of unsintered powder is commonly found in laser powder bed processes and is associated with lack of fusion [23,24]. In extreme cases, a network of lack of fusion defects cluster and connect to the free surface, creating tunneling porosity which is particularly detrimental to mechanical properties, even in the monolithic samples. However, the dominance of tunneling porosity was found to increase with decreasing strut size, presumably because smaller networks occupy a larger fraction of the overall cross-section.

The fracture surfaces in Fig. 12 compare representative horizontal



Fig. 7. Examples of strut failure locations for each combination of size and orientation (same samples as Fig. 4).

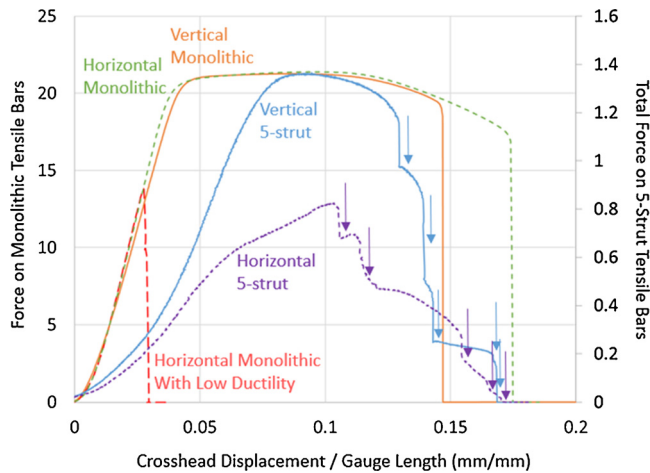


Fig. 8. Representative force-displacement curves for different sample categories. These representative examples were all of intermediate size (0.65 mm struts; 4.65 mm monolithic tensile bars).

and vertical struts. Vertical samples exhibited a fracture surface that was closer to what would be expected from ideal wrought samples: surface roughness played only a minor role and ductile dimples were present uniformly throughout the fracture surface. In contrast, the horizontal samples tended to have much smaller regions of ductile dimple fracture with lack of fusion and dripping roughness defects often reducing the true load-bearing area by up to ~50%, generally consistent with observed reductions in ultimate strength.

4. Discussion

The most important observation of the present study is the substantial deviation between nominal strut performance and observed strut performance. When compared to the larger, ‘monolithic’ tensile bars which typically meet the AMS 5604 minimum tensile strength of 930 MPa, the struts exhibited an average tensile strength of only 843 MPa with a substantial standard deviation of 146 MPa. Some of the worst struts exhibited a tensile strength of only ~400 MPa, or 40% of the monolithic value. These low values are particularly concerning for lattice integrity, since lattice failure is highly sensitive to weakest links – failed struts tend to drive local instability, causing a percolating propagation through more highly stressed neighboring struts as the stress redistributes.

This result has substantial implications for lattice design. One primary motivation for some lattice topologies such as the octet truss is an increased strength-to-weight ratio compared to that achievable with fully dense material. However, those calculations of improved performance are often based on nominal material properties. Almost invariably, lattice design (e.g. [3,25,26]) makes four assumptions that are called into question in the present study: (1) struts are ideally

Table 3

Average and standard deviation of failure strain (total elongation).

		Strain at Failure	Standard Deviation
Monolithic	All Monolithic	.108	.058
	All Monolithic (Excluding Low-ductility)	.131	.035
	Vertical	.115	.020
	Horizontal	.142	.040
	(Excluding Low-ductility)		
Struts	All Struts	.102	.025
	Vertical	.104	.020
	Horizontal	.100	.030

cylindrical in shape, (2) strut diameter is nominal, (3) intrinsic strut material properties are comparable to wrought or at least large-scale tensile properties for the printing process, and (4) all struts throughout the lattice are homogeneous with the same shape/properties independent of strut size or orientation. To resolve these issues, it will be necessary to incorporate realistic, stochastic properties into lattice design and shape optimization. The present study illustrates the importance of evaluating the effective stochastic properties of struts through strut-level characterization.

It is important to distinguish the present study from prior studies which have investigated the behavior of a single isolated strut [14]. The measurement of a single strut is somewhat analogous to the present study’s measurement of monolithic tensile response, although the comparatively large monolithic tensile bars in the present study reduce the relative influence of surface roughness and hence provide a more direct measure of the intrinsic tensile behavior. By contrast, when measuring the tensile performance of a 5-strut specimen, the load is divided among all five members. Load bearing is not necessarily uniformly subdivided among struts, due to their individual imperfections, as described in more detail in Section 4.2.

4.1. Roughness and orientation effects on heterogeneous response

The primary source for mechanical performance deviation is the non-ideal shape and size of the true load-bearing cross-section in the as-printed struts, including both the external profile of the sample and internal porosity. To first order, the as-printed surface roughness may be useful as a surrogate to infer the deviation from nominal strength. As strut diameter decreases, the ratio of surface roughness to gauge size increases, hence elevating the importance of roughness on apparent strength. Fig. 13 compares the ultimate strength of each sample to a measure of surface roughness. Four R_{rms} roughness measurements were calculated along each strut in MATLAB from slices of CT data, and the maximum of the twenty values for each sample is shown in the figure. Several other roughness metrics, such as the mean roughness, showed similar correlations with strength data, although from a causality perspective, maximum roughness arguably should have the most

Table 2

Average and standard deviation of ultimate tensile strength.

		Nominal Stress		Actual Stress	
		Average Ultimate Stress	Standard Deviation	Average Ultimate Stress	Standard Deviation
Monolithic	All Monolithic	874 MPa	273 MPa	904 MPa	280 MPa
	All Monolithic (Excluding Low-ductility)	980 MPa	38 MPa	1012 MPa	21 MPa
	Vertical	978 MPa	4 MPa	1002 MPa	21 MPa
	Horizontal	982 MPa	50 MPa	1019 MPa	20 MPa
	(Excluding Low-ductility)				
Struts	All Struts	656 MPa	180 MPa	843 MPa	146 MPa
	Vertical	792 MPa	60 MPa	924 MPa	59 MPa
	Horizontal	501 MPa	139 MPa	751 MPa	161 MPa

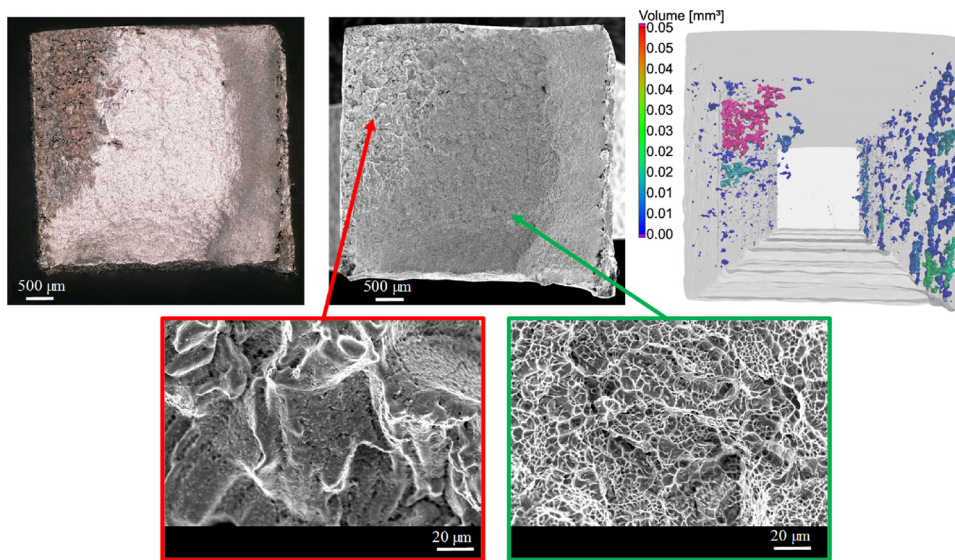


Fig. 9. Optical microscopy (top-left) showed a discoloured oxidized region in the top left corner corresponding to ~20% of the overall fracture surface. The presence of the shear lip on the right side of this fracture surface indicates that the crack ran from left to right, originating in the discoloured region. Scanning electron microscopy (top-middle and lower images) illustrate the morphological transition from a solidified free surface to a conventional ductile dimple failure. In the top-right image, the as-fabricated porosity in the tensile bar (coloured by pore volume) was imaged by computed tomography. This view looks down the long axis of the tensile bar, with the outer surface of the tensile bar shaded grey.

significant influence on mechanical performance. When comparing ultimate strength to strut roughness, the data is separated into two sections. From this figure, it is clear that the vertical struts possess lower surface roughness corresponding to higher average strength. Furthermore, the cluster size also changes consistently: the vertical samples show a much narrower distribution of roughness values and correspondingly a narrower range of strength values compared to the horizontal struts.

The data in Fig. 13 is also categorized by strut size where the different colours represent the different strut diameters. For the vertical samples, there does not appear to be a separation based on strut diameter. However, for horizontal struts, the larger struts (0.82 mm in green triangles) generally had higher strength than the smaller struts (0.50 mm in blue triangles). This is consistent with the concept that as the cross-sectional area increases, the strut's skin (outermost region of the cross-sectional area) is less impactful to the mechanical response. When the cross-sectional area is far larger than the skin, as in the monolithic samples, internal defects generally dominate part

performance and surface roughness is not a first-order contributor.

These results have implications for both lattice design and process optimization. With regard to design, a strategy for optimization will need to account for the substantial orientation effects. The directionality of the printing process produces substantially different material quality for different strut orientations, affecting optimization. While there have been numerous prior studies on building orientation effects relative to mechanical performance [27–29], those have predominantly focused on the role of microstructure or internal porosity on machined tensile bars. The present study emphasizes the role of build orientation on surface integrity, especially in the context of slender features printed above the build tray. With regard to process optimization, there may be an opportunity to mitigate these issues by adjusting the laser processing conditions (power, speed, etc.) depending on the local feature orientation: horizontal features may need a different scan strategy to achieve better surface properties. Slicing software that can adjust for local feature demands could help compensate for the observed effects in the present study.

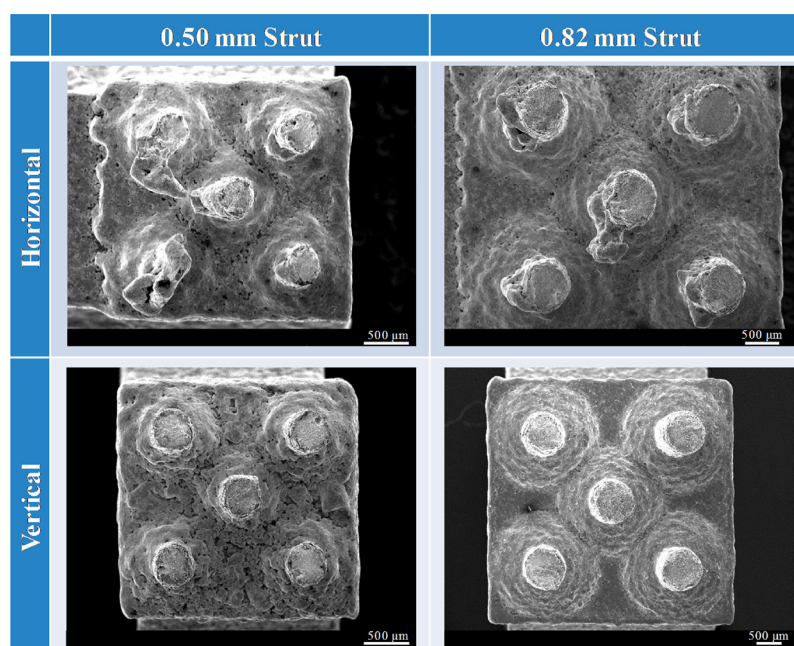


Fig. 10. Representative SEM fractography images.

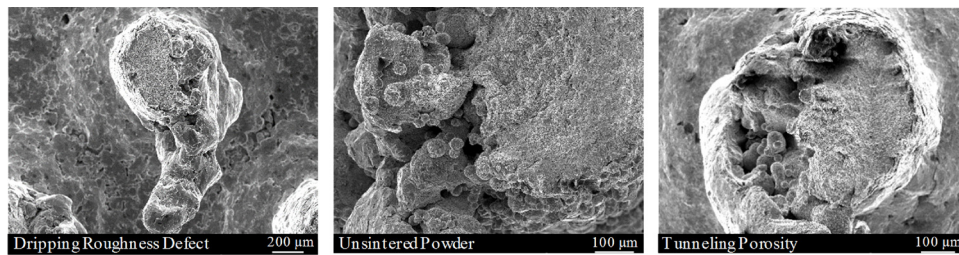


Fig. 11. Common heterogeneities in strut samples: Dripping roughness, unsintered powder, and tunneling porosity.

4.2. Structural mechanics analysis of CT data

To further interpret the experimental results, a series of simulations provided understanding of the impact of as-measured surface topography on the stochastic deviations in strut behavior. While the surface roughness value may provide a simple first-order metric for strut performance, detailed structural mechanics simulations can help quantify the role of inhomogeneous surface topography on mechanical performance. Due to the complex inhomogeneities of both the surface and internal defects of the struts, detailed information about the as-built structure of the struts is needed. The CT data provides this information, and with careful segmentation of the scan volume an accurate representation of each individual strut sample is computationally accessible.

The samples in Fig. 4 were used as a representative example dataset to analyze for each of the six strut conditions (two orientations and three strut sizes). The resulting simulations of von Mises stress distribution are shown in Fig. 14. In these plots, the local von Mises stress is normalized by the nominal applied stress. This nominal stress is the homogeneous stress value that would exist in a perfect strut of nominal dimensions, as determined by dividing the applied force by the nominal cross-sectional area of the as-designed strut. Since a value of 1 corresponds to an ideal strut, the upper limit of the scale colour bar indicates the degree of local stress concentration that occurs in different strut topologies. The current study did not attempt to validate the simulation method nor did it attempt to account for microstructural variations, but was merely intended to visualize the approximate magnitude of stress-concentrating effects caused by surface roughness. The precise quantitative values are not critical for this study. For this reason, stress values are reported in relative terms and with only 1-digit precision.

These visualizations are helpful to interpret the experimentally-observed mechanical performance of struts. Most obviously, all struts have some degree of geometric imperfection that gives rise to local stress concentrations. For all strut sizes, these imperfections are much more evident in the horizontally-oriented struts compared to the vertical struts. Also, for the horizontal struts, the smaller struts had more pronounced stress concentrations, as the topographic roughness was a larger fraction of the strut diameter. Beyond local variations in stress,

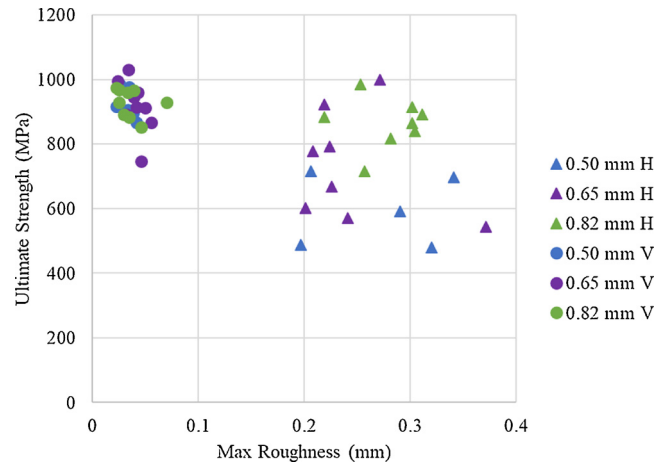


Fig. 13. Ultimate strength of strut samples versus the maximum roughness values.

there is also a relevant observation on the average stress in the struts. If the struts were ideal and of nominal diameter, the colour throughout the specimen would correspond to 1 on the colour bar. However, generally the small struts were undersized, leading to actual stresses that were elevated above the expected nominal stress. While the simulations are only for a single strut specimen for each condition, the stochastic topographic heterogeneities are expected to differ from specimen to specimen, giving rise to the large degree of scatter in specimen performance that was observed experimentally. Finally, as a disclaimer, these simulations are merely for linear elastic stresses and for the initial as-printed shape. They can be used to imply the stress state at the onset of yielding. After yielding, stresses will plastically redistribute, diminishing the local stress concentrating effects. Nevertheless, these simulations highlight the role that topographic heterogeneities play to dictate the mechanical response, and the trends shown in these analyses are at least qualitatively consistent with the experimentally-observed trends.

The present study focused only on the tensile performance of strut-

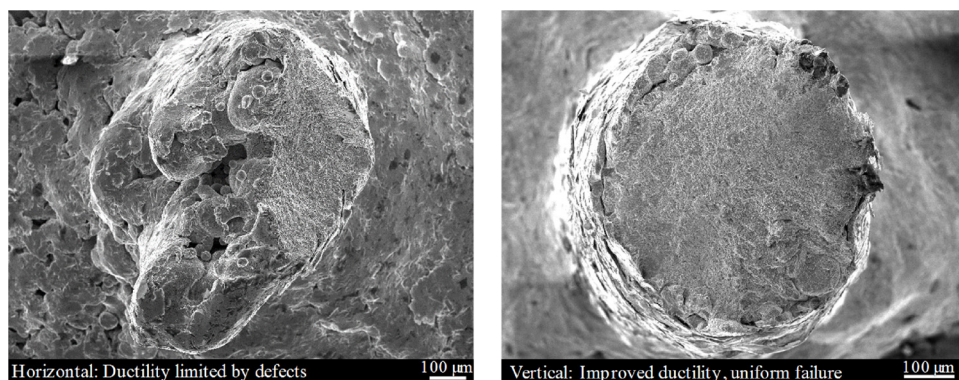


Fig. 12. Representative examples of degree of ductile failure uniformity.

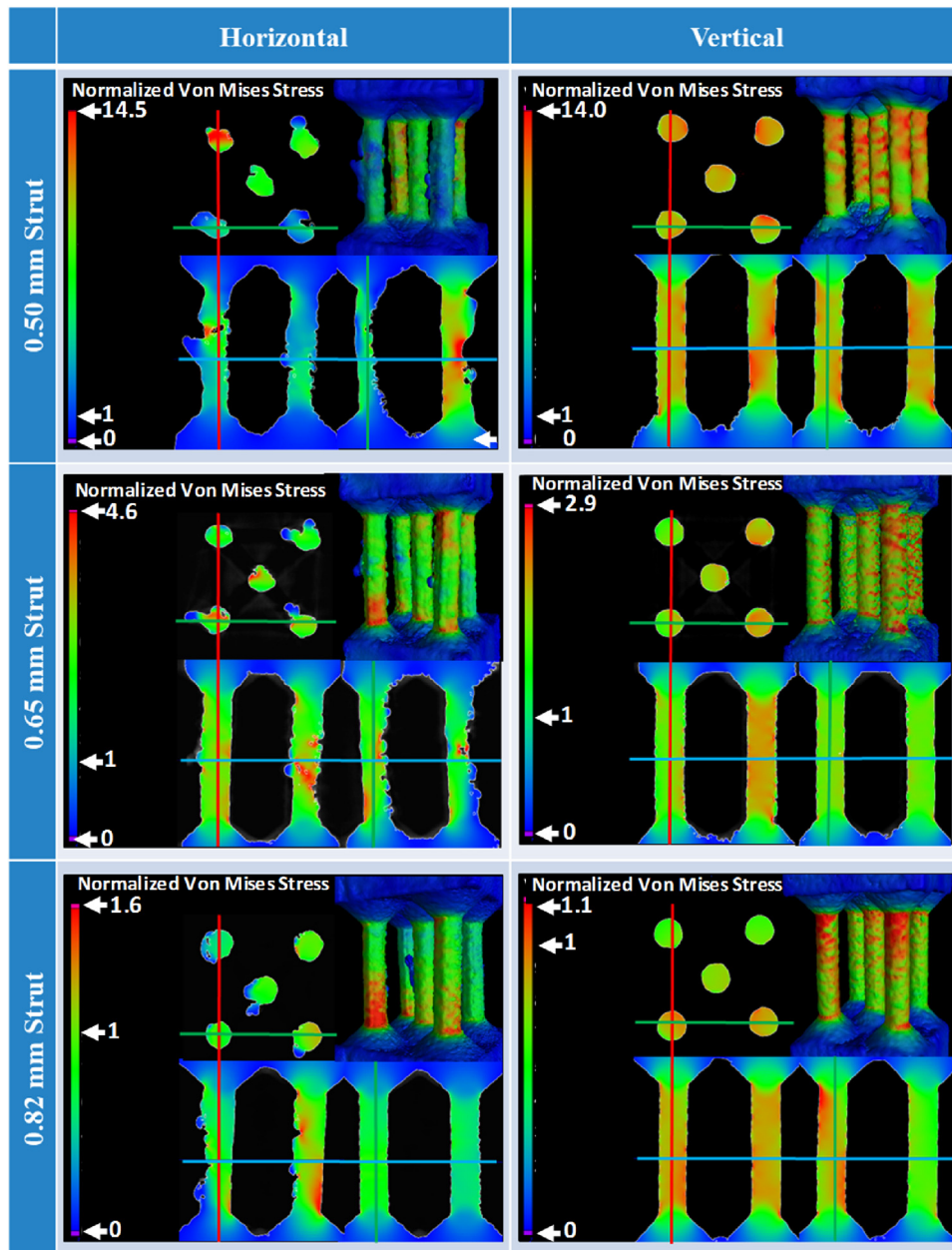


Fig. 14. Representative examples of stress distributions associated with actual strut shape. Scale bars show local von Mises stress values normalized by the nominal applied stress for a perfect strut.

like members. However, in addition to stretch-dominated struts, many lattices employ bending, compression, and buckling deformation modes. While the stretch-dominated mode likely is most critical in controlling failure in unit cells such as the octet truss, it would be beneficial to also examine the role that heterogeneities play on other relevant loading modes. Also, other strut properties of interest may include fracture toughness, fatigue behavior, or even creep behavior depending on the requirements of the application. Likewise, it may be relevant to study the effect of lattice heterogeneities in other relevant environments beyond room temperature quasi-static behavior, such as dynamic shock loading or elevated temperatures.

Variability in material performance has already been identified as a key concern for additively manufactured components. However, the variation in effective material properties becomes more significant at the finer feature sizes typically associated with lattices. The assumption of uniform unit cell performance will likely lead to non-conservative design, especially for weakest-link phenomena such as fatigue and

fracture. There are numerous stochastic approaches to design optimization which are the subject of research and commercialization, although inevitably there are two core challenges: (1) proper characterization of the heterogeneous performance (length-scale dependent property distributions), and (2) computational efficiency / tractability of the stochastic design process.

Inspection of lattices for undesirable heterogeneities will likely be an important challenge moving forward. Structures with many thousands of individual struts, some of which are deeply buried and difficult to observe even with tomographic methods, will likely require conservative design in the foreseeable future. Ultimately, it may be beneficial to address unit cell variability with design of protective features within the unit cell architecture. Periodic displacement-limiting features could serve to arrest cracks associated with weakest links and serve to truncate the detrimental effects of weakest links.

5. Conclusions

The present study highlights the important deviation in mechanical performance from the nominal expected performance due to the presence of local heterogeneities in the struts that serve as building blocks for lattice structures. Strut tensile specimens were used to enable efficient, high throughput quasi-static testing of struts and investigate the influence of strut heterogeneities. When compared against a control group of larger scale tensile bars printed on the same build plate, the strut features possessed mechanical properties that were considerably lower on average and much more inconsistent from strut-to-strut. There are several sources that were evaluated for this discrepancy: deviations from the nominal cylindrical shape, deviation in diameter from the expected value, and the presence of several defect types including lack-of-fusion porosity and surface-connected tunneling (networked) porosity. Defects were particularly pronounced in horizontally-oriented struts, where the lack of support from underlying solid material even resulted in a form of dripping defect. As a result, horizontal struts showed even lower strength values and higher scatter compared to the vertical struts. While the study placed this result in the context of lattice geometries which tend to rely on many fine scale features, the results are generally applicable to all geometries that contain structural features approaching the scale of surface topography in additive manufacturing processes.

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References

- [1] M. Lapine, et al., Structural tunability in metamaterials, *Appl. Phys. Lett.* 95 (8) (2009) 084105.

- [2] L.R. Meza, et al., Resilient 3D hierarchical architected metamaterials, *Proc. Natl. Acad. Sci.* 112 (37) (2015) 11502–11507.
- [3] J.-S. Liu, T.J. Lu, Multi-objective and multi-loading optimization of ultralightweight truss materials, *Int. J. Solids Struct.* 41 (3–4) (2004) 619–635.
- [4] L. Dong, V. Deshpande, H. Wadley, Mechanical response of Ti–6Al–4V octet-truss lattice structures, *Int. J. Solids Struct.* 60 (2015) 107–124.
- [5] K. Bertoldi, et al., Negative Poisson's ratio behavior induced by an elastic instability, *Adv. Mater.* 22 (3) (2010) 361–366.
- [6] K.E. Evans, A. Alderson, Auxetic materials: functional materials and structures from lateral thinking!, *Adv. Mater.* 12 (9) (2000) 617–628.
- [7] Buckminster, F.R., *Synergetic building construction*. 1961, Google Patents.
- [8] X. Zheng, et al., Ultralight, ultrastiff mechanical metamaterials, *Science* 344 (6190) (2014) 1373–1377.
- [9] C. Körner, A. Bauereiß, E. Attar, Fundamental consolidation mechanisms during selective beam melting of powders, *Model. Simul. Mat. Sci. Eng.* 21 (8) (2013) 085011.
- [10] S.P. Moylan, et al., Lessons Learned in Establishing the NIST Metal Additive Manufacturing Laboratory, NIST Technical Note (2013).
- [11] M. O'Masta, et al., The fracture toughness of octet-truss lattices, *J. Mech. Phys. Solids* 98 (2017) 271–289.
- [12] F. Mitri, F. Garzon, D. Sinha, Characterization of acoustically engineered polymer nanocomposite metamaterials using x-ray microcomputed tomography, *Rev. Sci. Instrum.* 82 (3) (2011) 034903.
- [13] A. Maiti, et al., 3D printed cellular solid outperforms traditional stochastic foam in long-term mechanical response, *Sci. Rep.* 6 (2016) 24871.
- [14] R. Gümruk, R. Mines, Compressive behaviour of stainless steel micro-lattice structures, *Int. J. Mech. Sci.* 68 (2013) 125–139.
- [15] V. Weißmann, et al., Comparison of single Ti6Al4V struts made using selective laser melting and electron beam melting subject to part orientation, *Metals* 7 (3) (2017) 91.
- [16] J. Zhou, P. Shrotriya, W. Soboyejo, On the deformation of aluminum lattice block structures: from struts to structures, *Mech. Mater.* 36 (8) (2004) 723–737.
- [17] M. Leary, et al., Selective laser melting (SLM) of AlSi12Mg lattice structures, *Mater. Des.* 98 (2016) 344–357.
- [18] M.J. Milroy, C. Bradley, G.W. Vickers, Segmentation of a wrap-around model using an active contour, *Computer-Aided Des.* 29 (4) (1997) 299–320.
- [19] D. Boffi, L. Gastaldi, A finite element approach for the immersed boundary method, *Comput. Struct.* 81 (8–11) (2003) 491–501.
- [20] V.G. GmbH, Germany, VGStudio MAX 3.0 Reference Manual, (2016).
- [21] O. Cansizoglu, et al., Properties of Ti–6Al–4V non-stochastic lattice structures fabricated via electron beam melting, *Mater. Sci. Eng. A* 492 (1–2) (2008) 468–474.
- [22] B.L. Boyce, et al., Extreme-value statistics reveal rare failure-critical defects in additive manufacturing, *Adv. Eng. Mater.* 19 (8) (2017).
- [23] W.E. Frazier, Metal additive manufacturing: a review, *J. Mater. Eng. Perform.* 23 (6) (2014) 1917–1928.
- [24] H. Gong, et al., Analysis of defect generation in Ti–6Al–4V parts made using powder bed fusion additive manufacturing processes, *Addit. Manuf.* 1 (2014) 87–98.
- [25] A. Clausen, et al., Topology optimized architectures with programmable Poisson's ratio over large deformations, *Adv. Mater.* 27 (37) (2015) 5523–5527.
- [26] V.S. Deshpande, N.A. Fleck, M.F. Ashby, Effective properties of the octet-truss lattice material, *J. Mech. Phys. Solids* 49 (8) (2001) 1747–1769.
- [27] B.E. Carroll, T.A. Palmer, A.M. Beese, Anisotropic tensile behavior of Ti–6Al–4V components fabricated with directed energy deposition additive manufacturing, *Acta Mater.* 87 (2015) 309–320.
- [28] M. Simonelli, Y.Y. Tse, C. Tuck, Effect of the build orientation on the mechanical properties and fracture modes of SLM Ti–6Al–4V, *Mater. Sci. Eng. A* 616 (2014) 1–11.
- [29] T.M. Mower, M.J. Long, Mechanical behavior of additive manufactured, powder-bed laser-fused materials, *Mater. Sci. Eng. A* 651 (2016) 198–213.